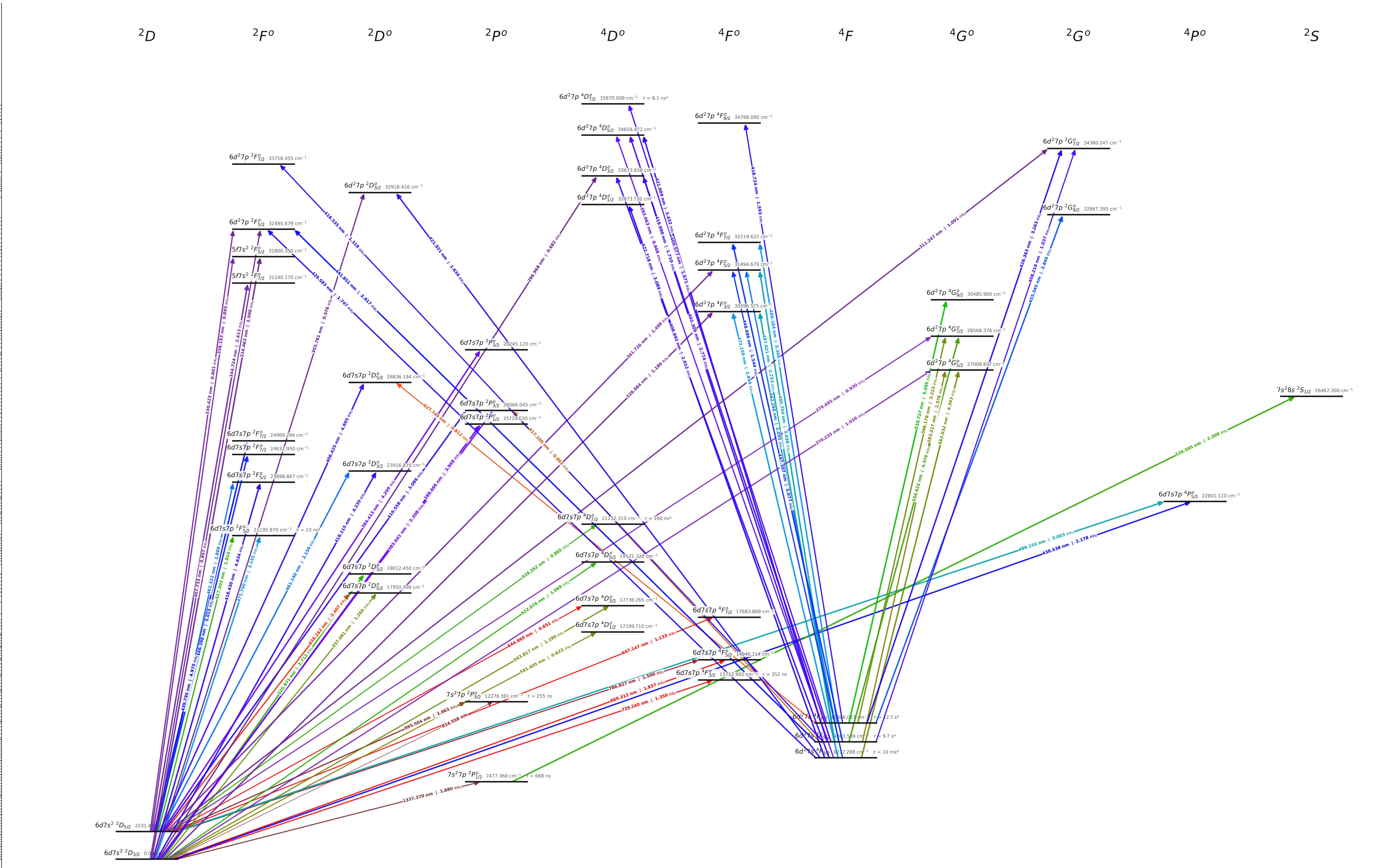


Actinium

Ac (Ac I). Strongest classified transitions below 2 μm (NIST): 45 levels, 72 lines.

Ac⁺ ionisation limit 43394.52 cm⁻¹ = 5.38023 eV (230.444 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
6d7s² ²D _{3/2} – 6d7s7p ²P _{3/2}	383.6409	383.5321	781.4404	2.308	4.780e+07	–	–	hyperfine structure of 227Ac measured
6d7s² ²D _{3/2} – 6d7s7p ²P _{1/2}	388.6660	388.5559	771.3369	3.948	2.690e+08	–	–	hyperfine structure of 227Ac measured
6d7s² ²D _{5/2} – 6d7s7p ²P _{3/2}	419.5579	419.4397	714.5437	3.086	6.530e+07	–	–	hyperfine structure of 227Ac measured
6d7s² ²D _{3/2} – 6d7s7p ²P _{5/2}	438.5378	438.4147	683.61821(10)	2.178	1.900e+07	–	–	the actinium laser-spectroscopy line: first step of the r...
6d7s² ²D _{3/2} – 6d7s7p ²F _{5/2}	439.7951	439.6716	681.6639	4.975	7.370e+07	–	–	hyperfine structure of 227Ac measured
6d7s² ²D _{3/2} – 6d7s7p ²F _{3/2}	729.2402	729.0394	411.1025	1.350	2.380e+06	0.4521	83.8	first-step line used for the lifetime measurement of the...
6d7s² ²D _{3/2} – 7s²7p ²P _{3/2}	814.5584	814.3345	368.0429	–	–	0.6241	–	first step of the two-step resonance ionization in which...
6d7s² ²D _{3/2} – 7s²7p ²P _{1/2}	1337.3704	1337.0048	224.1656	1.880	1.497e+06	0.2383	100	first observed 2020 by two-step resonance ionization

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000

227Ac hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
6d7s² ²D _{3/2}	50.5(5)	595.6(15)	F=0: +555.27 F=1: +10.13 F=2: -484.55 F=3: +262.44
6d7s² ²D _{5/2}	255(4)	706(19)	F=1: -844.55 F=2: -899.35 F=3: -452.05 F=4: +1132.75
7s²7p ²P _{3/2}	513(8)	1260(52)	F=0: -348.75 F=1: -1095.75 F=2: -1329.75 F=3: +1469.25
6d7s7p ²P _{3/2}	2104.8(22)	110.1(22)	F=1: -10973.13 F=2: -6851.61 F=3: -586.75 F=4: +7920.52
6d7s7p ²F _{5/2}	-280(4)	1278(30)	F=2: +2576.40 F=3: +823.00 F=4: -663.20 F=5: -1152.20
6d7s7p ²P _{1/2}	1972(4)	–	F=1: -2464.38 F=2: +1478.62
6d7s7p ²P _{3/2}	-990.7(12)	-21(4)	F=0: +3689.25 F=1: +2719.25 F=2: +758.55 F=3: -2234.25

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
6d7s² ²D _{3/2} – 6d7s7p ²P _{3/2}	438.538	225-215	-50300(210)
6d7s² ²D _{3/2} – 6d7s7p ²P _{1/2}	438.538	226-215	-54700(220)
6d7s² ²D _{3/2} – 6d7s7p ²P _{5/2}	438.538	227-215	-58300(100)
6d7s² ²D _{3/2} – 6d7s7p ²P _{3/2}	438.538	228-215	-62000(1000)
6d7s² ²D _{3/2} – 6d7s7p ²P _{5/2}	438.538	229-215	-64900(320)
6d7s² ²D _{3/2} – 6d7s7p ²P _{3/2}	438.538	214-215	2969(42)
6d7s² ²D _{3/2} – 6d7s7p ²P _{5/2}	438.538	213-215	4282(404)
6d7s² ²D _{3/2} – 6d7s7p ²P _{1/2}	438.538	212-215	7710(393)

Shift = ν(first isotope) – ν(second isotope).